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```
clear;
clc;
close all;
```

Load all three Talley test files

```
sparkData = load('SPARK_ONLY_001.mat');
noiseData = load('WN_ONLY_001.mat');
mixedData = load('SPARK_NOISE_EQUAL_RMS_001.mat');

sparkIQ = sparkData.eventIQ;
noiseIQ = noiseData.eventIQ;
mixedIQ = mixedData.eventIQ;

fs = sparkData.fs;
fc = sparkData.fc;

fprintf('Sample rate: %.0f Hz\n', fs);
fprintf('Center frequency: %.1f MHz\n\n', fc/1e6);

Sample rate: 2400000 Hz
Center frequency: 145.9 MHz
```

Magnitude

```
sparkMag = abs(sparkIQ);
noiseMag = abs(noiseIQ);
mixedMag = abs(mixedIQ);
```

Measurements

```
sparkRMS = rms(sparkMag);
sparkPeak = max(sparkMag);
sparkPeakRMS = sparkPeak / sparkRMS;

noiseRMS = rms(noiseMag);
noisePeak = max(noiseMag);
```

```
noisePeakRMS = noisePeak / noiseRMS;
```

```
mixedRMS = rms(mixedMag);  
mixedPeak = max(mixedMag);  
mixedPeakRMS = mixedPeak / mixedRMS;
```

Print results

```
fprintf('=====  
SPARK ONLY  
=====\n');  
fprintf('RMS magnitude: %.6f\n', sparkRMS);  
fprintf('Peak magnitude: %.6f\n', sparkPeak);  
fprintf('Peak / RMS ratio: %.2f\n\n', sparkPeakRMS);
```

```
fprintf('=====  
WHITE NOISE ONLY  
=====\n');  
fprintf('RMS magnitude: %.6f\n', noiseRMS);  
fprintf('Peak magnitude: %.6f\n', noisePeak);  
fprintf('Peak / RMS ratio: %.2f\n\n', noisePeakRMS);
```

```
fprintf('=====  
SPARK + NOISE  
=====\n');  
fprintf('RMS magnitude: %.6f\n', mixedRMS);  
fprintf('Peak magnitude: %.6f\n', mixedPeak);  
fprintf('Peak / RMS ratio: %.2f\n\n', mixedPeakRMS);
```

```
=====  
SPARK ONLY  
=====  
RMS magnitude: 0.008422  
Peak magnitude: 0.251189  
Peak / RMS ratio: 29.83
```

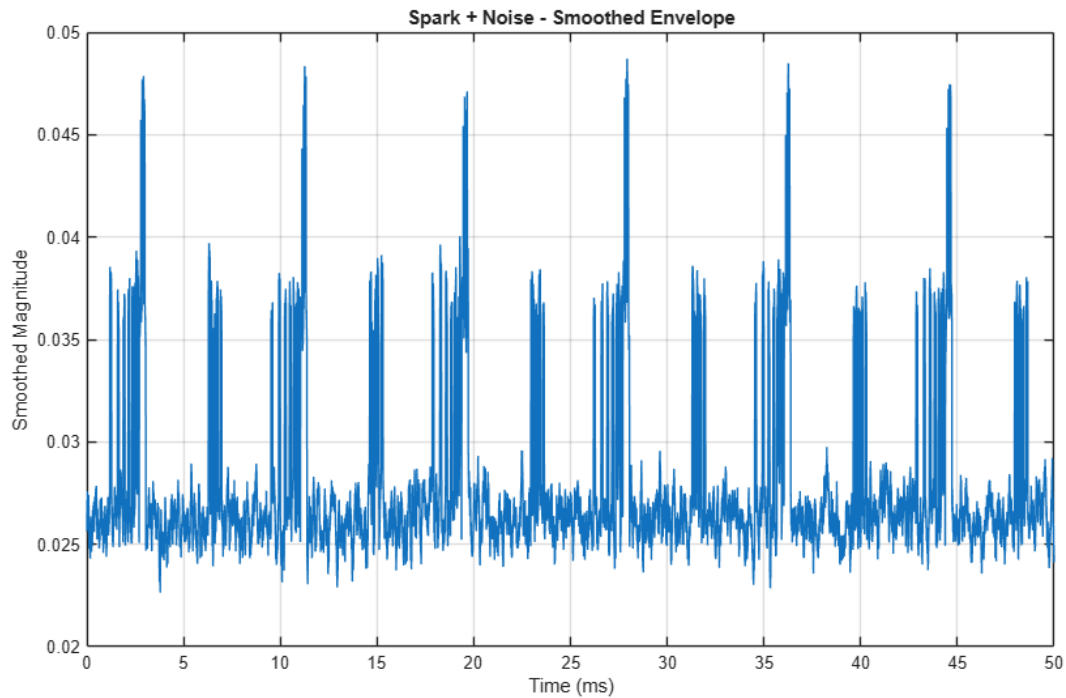
```
=====  
WHITE NOISE ONLY  
=====  
RMS magnitude: 0.017786  
Peak magnitude: 0.071870  
Peak / RMS ratio: 4.04
```

```
=====  
SPARK + NOISE  
=====  
RMS magnitude: 0.041858  
Peak magnitude: 0.950000  
Peak / RMS ratio: 22.70
```

Plot first 50 ms

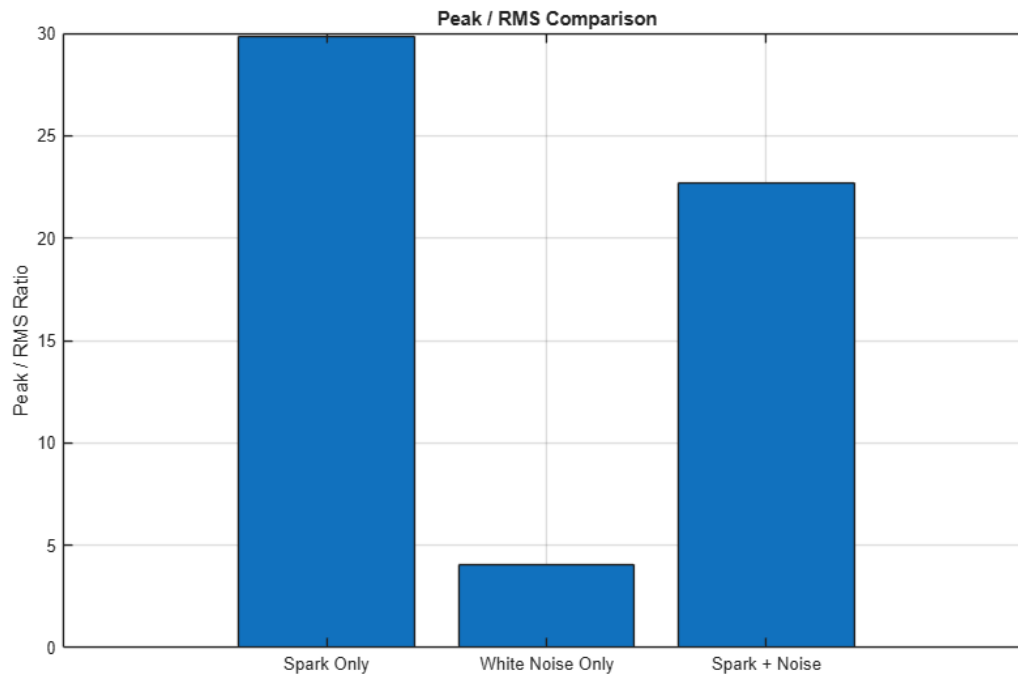
```
showTime = 0.05;  
Nshow = round(showTime * fs);  
t = (0:length(mixedIQ)-1) / fs;  
  
windowLength = 200;  
  
sparkSmooth = movmean(sparkMag, windowLength);  
noiseSmooth = movmean(noiseMag, windowLength);  
mixedSmooth = movmean(mixedMag, windowLength);  
  
figure;  
plot(t(1:Nshow)*1000, mixedSmooth(1:Nshow));
```

```
xlabel('Time (ms)');  
ylabel('Smoothed Magnitude');  
title('Spark + Noise - Smoothed Envelope');  
grid on;
```



Compare all three

```
figure;  
  
values = [sparkPeakRMS, noisePeakRMS, mixedPeakRMS];  
  
bar(values);  
  
set(gca, 'XTickLabel', ...  
    {'Spark Only', 'White Noise Only', 'Spark + Noise'});  
  
ylabel('Peak / RMS Ratio');  
title('Peak / RMS Comparison');  
grid on;
```



First simple detection rule

```
threshold = 8;

fprintf('===== SIMPLE DETECTOR =====\n');
fprintf('Threshold = %.2f\n', threshold);
fprintf('Mixed Peak/RMS = %.2f\n', mixedPeakRMS);

if mixedPeakRMS > threshold
    fprintf('CANDIDATE SPARK EVENT DETECTED\n');
else
    fprintf('NO EVENT DETECTED\n');
end

===== SIMPLE DETECTOR =====
Threshold = 8.00
Mixed Peak/RMS = 22.70
CANDIDATE SPARK EVENT DETECTED
```

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